

InvenIQ

AI-Powered Inventory Decision Intelligence for SMEs

Stop Guessing · Start Forecasting

Malaysian SMEs lose RM 14B/year to inventory blind spots.



73%

of SME retailers have no demand forecasting tool



RM 14B

lost annually to overstock and stockouts combined

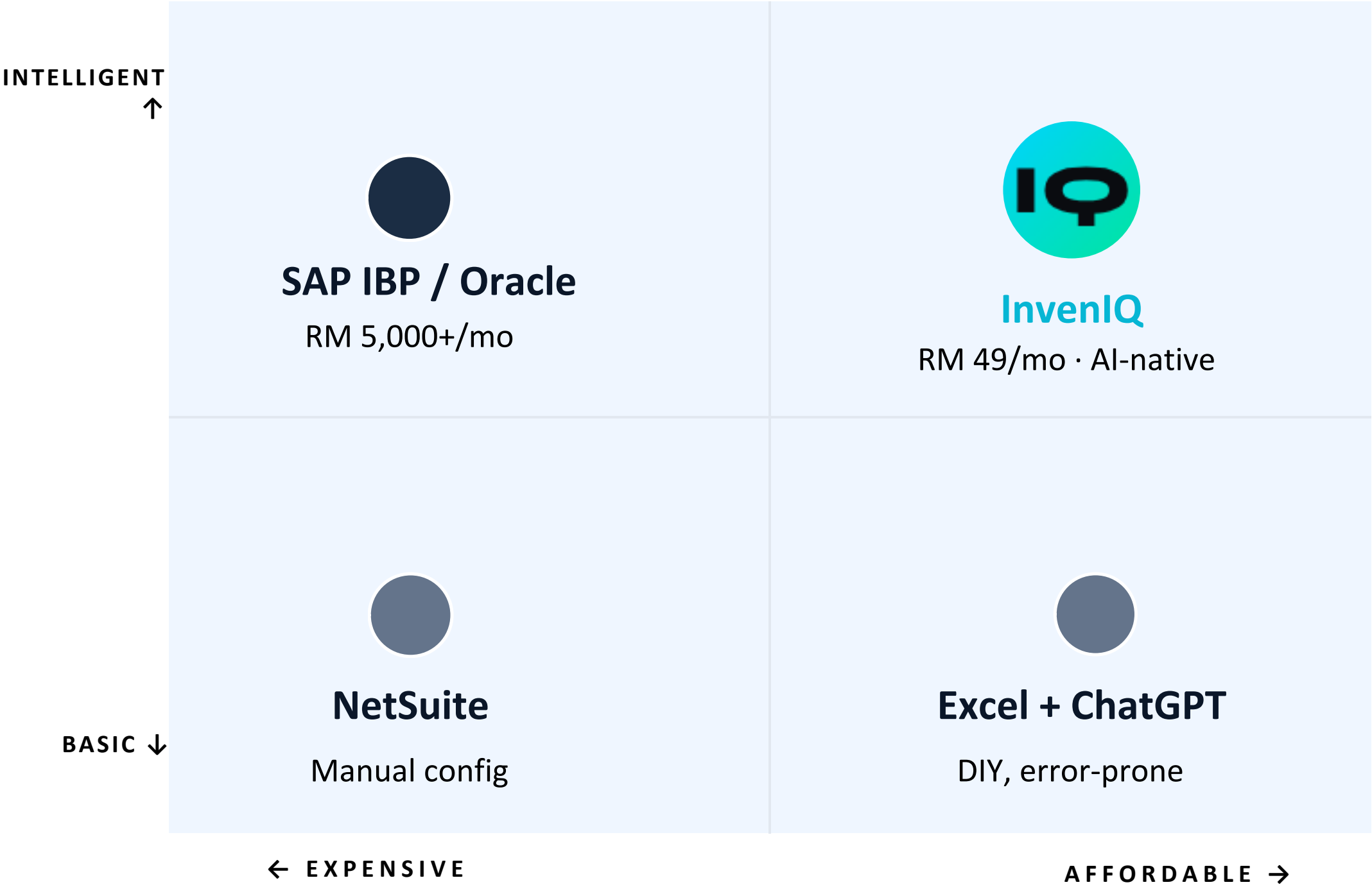


Overstock

Ties up vital cash flow in slow-moving items.

Sources: SME Bank MY 2025 · DOSM Retail Index 2025

The forecasting market has a shape problem.



WHAT WE SEE

The intelligent + affordable quadrant is empty.

Enterprise tools are too heavy. Generic AI chatbots can't read a CSV. SMEs are stuck choosing between 'expensive consultants' or 'good luck'.

InvenIQ fills the gap.

Why we win the SME segment?

	SAP IBP	Oracle NetSuite	Generic ChatGPT	InvenIQ
Setup time	3–6 months	4–8 weeks	Instant	Instant
Monthly cost	RM 5,000+	RM 1,500+	RM 100	RM 49
Multi-modal input	CSV only	CSV only	Text only	CSV + Image
Live signal fusion	Add-on module	Manual	None	Native + cached
MY local context	Generic	Generic	Partial	MY-tuned prompts

VERDICT: *We're not competing with SAP. We're competing with the Excel spreadsheet — and we win on every axis.*

An AI advisor that sees what your CSV can't.



01

Multi-Modal Ingestion

Drop a CSV, an Excel file, or a photo of a paper stock sheet. We read them all into a unified inventory model.



02

Live Signal Fusion

Weather, public holidays, news, and market trends — fused with your stock data in real time via dedicated APIs.



03

Conversational Forecasting

Ask in plain English: "What should I overstock for Hari Raya?" Get the answer, the reasoning, and a chart.

VALUE PROP: Enterprise-grade demand intelligence at hackathon-grade cost — designed for the SME who can't afford a data team.

Time to see it work.

→ **LIVE DEMO**

WHAT TO WATCH FOR

1

Multi-Modal Upload

Drop a CSV — instant header validation + preview.

2

External Signal Fusion

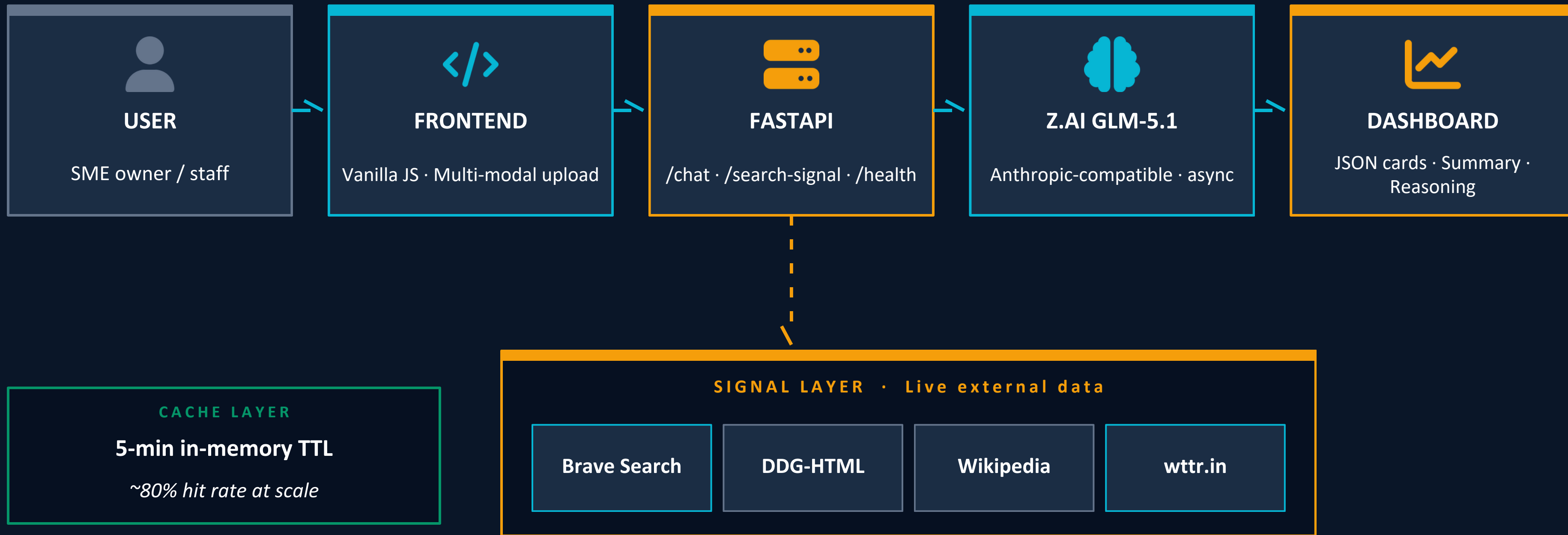
Click 'Weather' — live Malaysia forecast in 2 seconds.

3

Agentic Reasoning

Ask: "What should I stock more of?" — watch it think.

End-to-end data flow.



Hosted on Render · Stateless FastAPI · Async I/O · 60s timeout · Python 3.11

How a question becomes a forecast.

**~30s**

Avg end-to-end latency
p50, warm container

~minimal

Avg tokens per turn
input + output

100%

JSON contract conformance
zero parse failures since DEF-01 fix

Five providers. Zero single points of failure.

PROVIDER PRIORITY CHAIN

1°	DuckDuckGo HTML <i>Direct scrape · No key needed</i>	ALWAYS ON
2°	Wikipedia <i>Always works · Encyclopedic last resort</i>	ALWAYS ON
3°	SerpAPI (Google) <i>Paid fallback · Highest quality results</i>	OPT-IN
4°	Tavily API <i>AI-native · Built for agentic search</i>	OPT-IN
5°	Brave Search API <i>Free 2k/mo · Most reliable on cloud IPs</i>	OPT-IN

THE ENGINEERING SOLUTION

Cascading provider fallback.

Try the highest-quality available source first; fall through automatically on failure. No matter the cloud, the IP, or the API status — search always returns something.

→ */search-debug endpoint shows which provider answered each query.*

Hardened Infrastructure

Engineered for Security and Reliability



Zero-Secret Architecture Complete decoupling of sensitive credentials from source code using environment injection.



Fail-Safe Mechanisms Automated defensive checks prevent the system from executing in any unverified state.

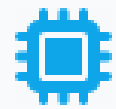


Enterprise Access Control Strict network allowlisting replaces open CORS policies to protect core API endpoints with JWT authentication.

Tracked as GitHub Issues #1, #2, #3 — searchable via labels: tech-debt, security, enhancement.

The InvenIQ Advantage

Proprietary architecture engineered for high-precision inventory intelligence.



Token Efficiency

Proprietary "Kitchen Sink" summarization engine (powered by generateDataSummary) processes 50k rows in-browser, slashing API costs and ensuring faster response times.



Zero-Hallucination Guardrails

the system prompt hard-codes a rule: the AI can only reference Product IDs that exist in your uploaded CSV. It literally cannot invent data.



Agentic UI Orchestration

Orchestrates real-time JSON payloads (addStructuredResponse()) to render dynamic, color-coded dashboard cards instead of standard text-heavy chat logs.



Resilient Architecture

Decoupled serverless backend on Render with 5-minute TTL signal caching(_SIGNAL_CACHE_TTL = 300.0) for high availability and industry-standard scalability.

Efficiency, Security and CI/CD



0

secrets in source

- ✓ API key in env vars (DEF-05)
- ✓ Old key revoked + rotated
- ✓ .env.example committed
- ✓ RuntimeError on missing env
- ✓ CORS hardening tracked (TD-01)



<90s

push-to-live

- ✓ GitHub Actions on push
- ✓ py_compile + pytest + smoke
- ✓ Render auto-deploy on website
- ✓ Zero manual ops per release
- ✓ /health endpoint for monitors



RM 0.04

per user session

- ✓ Z.AI flat-rate plan (\$18/mo)
- ✓ 5-min signal cache (~80% hit)
- ✓ Decoupled signal fetch
- ✓ Client-side CSV summary
- ✓ max_tokens cap = no runaways

Three tiers. One mission: forecasting for everyone.

Starter

Free

Forever

- ✓ Basic stock tracking
- ✓ Limited CSV uploads (5/mo)
- ✓ Manual signal fetching
- ✓ Community support

MOST POPULAR

Professional

RM 49

/month

- ✓ Full GLM-5.1 reasoning
- ✓ Unlimited CSV uploads
- ✓ Auto signal fusion
- ✓ Multi-modal input (image/CSV)
- ✓ Priority email support

Enterprise

Custom

Contact us

- ✓ POS API integration
- ✓ Custom dashboards & reports
- ✓ On-prem deployment option
- ✓ SLA + dedicated CSM

Wedge into MY mini-marts. Expand to SEA SMEs.

PHASED ROLLOUT

- 1

Q3 26

Launch & Wedge

100 mini-marts in Klang Valley · free tier
- 2

Q1 27

Monetize & Localize

Pro tier · MY festival pack · BM language
- 3

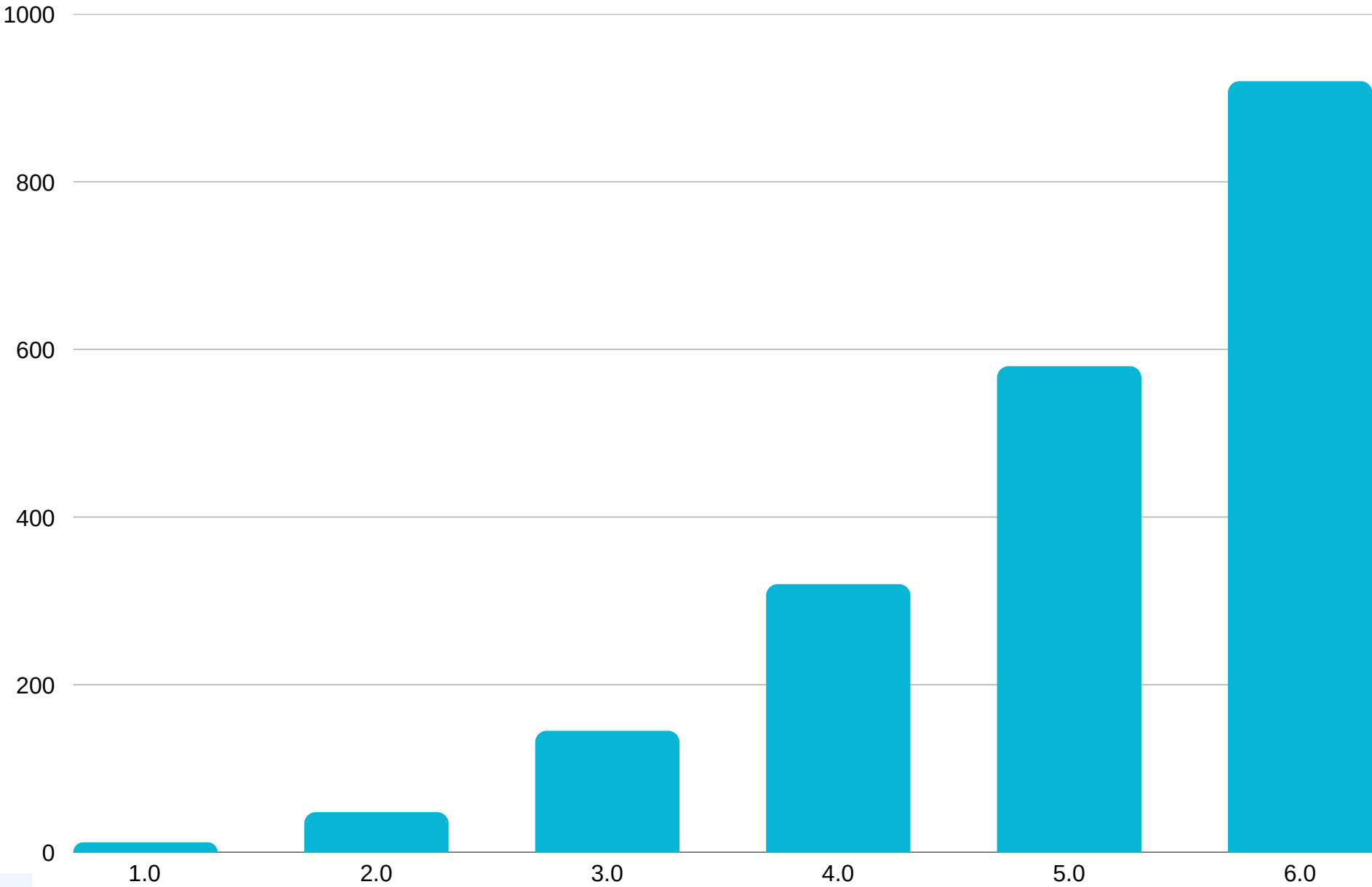
Q4 27

Regional Expansion

ID, TH, VN · partner with regional POS vendors

BREAK-EVEN Q2 2027 · TARGET LTV/CAC 5.2×

ARR PROJECTION (RM '000)



THANK YOU

Stock smarter.

Stock kinder to your cashflow.



TRY IT LIVE

z-ai-7zxw.onrender.com



READ THE CODE

<https://github.com/andrewteeky-code/UM-Hackathon-2026>



ASK ANYTHING

Q&A starts now

Team OREO · UMHackathon 2026 · Powered by Z.AI GLM-5.1



INVENTORY INTELLIGENCE PLATFORM

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